

Effects of Illegal Logging and Deforestation in the Sierra Madre, Philippines

The **Sierra Madre Mountain Range** is one of the Philippines' most important forest ecosystems. Stretching roughly **540 kilometers across eastern Luzon**, it covers or influences parts of several provinces and provides forests, watersheds, wildlife habitat, and natural protection against tropical cyclones. The Climate Change Commission describes it as a critical natural defense for Luzon, while studies identify the **Northern Sierra Madre Natural Park** as one of the country's most important protected areas for biodiversity.

 Climate Chang...+1

Illegal logging and other forms of forest degradation threaten these functions. The effects are not limited to trees: they can affect **wildlife, water supplies, agriculture, communities, disaster risk, climate regulation, and local livelihoods**.

1. Loss of biodiversity and wildlife habitat

One of the most serious effects is the destruction and fragmentation of natural habitats.

Sierra Madre forests support numerous plant and animal species, including species that occur only in the Philippines. When trees are illegally cut, wildlife loses nesting sites, food sources, shelter, and breeding areas. As forests become fragmented, populations can also become isolated, making it more difficult for animals to move between suitable habitats.

Research on the Northern Sierra Madre Natural Park identifies illegal logging as a direct threat to both **biodiversity and rural livelihoods**. More recent research on illegal forest activities in the southeastern Sierra Madre likewise reports that illegal logging, slash-and-burn farming, and agricultural expansion threaten plant and animal species and conservation efforts.

 JST...+1

Website-friendly impact:

Illegal logging destroys and fragments wildlife habitats, reducing the ability of native and endemic species to survive and reproduce.

2. Increased soil erosion

Trees and other vegetation help stabilize soil through their roots. Forest canopies also reduce the direct impact of heavy rainfall on the ground.

When forests are cleared, exposed soil is more easily washed away during heavy rain. The problem can become particularly severe in the steep terrain of the Sierra Madre.

Deforestation in the Philippines has been associated with **soil erosion, landslides, flooding, biodiversity loss, and watershed degradation.**  [M...](#)

Website-friendly impact:

Removing trees exposes mountain soil to heavy rainfall, increasing soil erosion and making slopes more vulnerable to degradation.

3. Higher risk of landslides

The Sierra Madre contains steep mountainous terrain and is regularly exposed to intense rainfall and tropical cyclones. Forest degradation can make already vulnerable slopes more susceptible to landslides.

Vegetation helps hold soil together and contributes to water absorption. When vegetation is removed, rainfall can produce faster surface runoff and destabilize exposed slopes.

Research concerning Sierra Madre watersheds notes that heavy rainfall can produce rapid runoff and landslides, particularly in **logged-over and denuded areas.**  [c.om.m unit y fore...](#)

Website-friendly impact:

Deforestation can increase slope instability, contributing to landslides during periods of heavy rainfall and typhoons.

4. Increased flooding in downstream communities

Healthy forests act like natural water-management systems. Forest vegetation and soils can absorb and temporarily retain rainfall, slowing the movement of water toward rivers and low-lying areas.

When forests are cleared, more rainwater can rapidly flow downhill. This can increase runoff, raise river levels, and transport large quantities of soil and sediment into waterways.

Research on Sierra Madre watersheds has documented how rapid runoff, erosion, river sedimentation, and the convergence of water from tributaries can contribute to **lo wla n d flooding.**  [c.om.m unit y fore...](#)

Website-friendly impact:

Forest loss reduces the landscape's ability to regulate rainfall runoff, potentially increasing flooding downstream.

5. Degradation of watersheds and water supplies

The Sierra Madre contains important watersheds that supply water for communities, agriculture, and other human activities.

Forests regulate the movement of water through the landscape. Deforestation can increase sedimentation in rivers and reservoirs while reducing the ability of watersheds to maintain stable water flows.

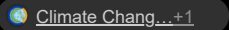
For example, research on the southeastern Sierra Madre describes protected forest areas as important sources of **fresh water for irrigation and industrial purposes** including hydroelectric power generation. 

Website-friendly impact:

Deforestation damages watersheds by increasing erosion and sedimentation and disrupting the natural regulation of water.

6. Reduced protection against typhoons

The Sierra Madre is particularly important because it lies along the eastern side of Luzon, where many tropical cyclones first encounter land.

Its forests and mountainous terrain provide a natural barrier that can help reduce the effects of storms before they reach communities farther inland. The Philippine Climate Change Commission describes the range as a **natural defense against powerful tropical cyclones**. 

When forest cover is degraded, however, communities can become more vulnerable to the secondary effects of intense rainfall, including flooding, erosion, and landslides.

Important distinction: The mountains themselves do not simply "stop" typhoons. Their terrain interacts with storms, while intact forests help protect soil, watersheds, and ecosystems from the resulting rainfall and runoff.

Website-friendly impact:

Forest degradation weakens the Sierra Madre's ability to provide ecosystem-based protection against the impacts of severe storms.

7. Increased sedimentation of rivers

When exposed soil is washed from deforested slopes, the sediment can enter streams and rivers.

Excess sediment can make waterways shallower and affect aquatic habitats. It can also reduce the capacity of waterways and reservoirs to store water.

In Sierra Madre watersheds, studies have connected forest degradation with **heavy soil erosion and sedimentation of riverbeds**. [community fore...](#)

Website-friendly impact:

Soil exposed by deforestation can be carried into rivers, increasing sedimentation and degrading freshwater ecosystems.

8. Loss of carbon storage and increased climate impacts

Forests store carbon in trees, vegetation, roots, and soils. When forests remain intact, they remove carbon dioxide from the atmosphere through photosynthesis and store it as biomass.

The Climate Change Commission identifies the Sierra Madre as an important **carbon sink**, with approximately 1.4 million hectares of forested area cited in its 2024 report.

[Climate Chang...](#)

When trees are cut and forests are degraded, this carbon-storage capacity is reduced. If forest biomass is burned or decomposes after clearing, some of the stored carbon can also return to the atmosphere as greenhouse gases.

Website-friendly impact:

Deforestation reduces the Sierra Madre's capacity to store carbon, weakening an important natural mechanism for climate regulation.

9. Damage to agriculture and rural livelihoods

The consequences of illegal logging can extend beyond the forest itself.

Many communities around the Sierra Madre depend on natural resources and surrounding ecosystems for agriculture, water, forest products, and other livelihood activities. Forest degradation can reduce the environmental resources on which these communities depend.

A study of illegal logging in the Northern Sierra Madre Natural Park found that illegal logging threatens **rural livelihoods** and can obstruct sustainable rural development by damaging ecosystems and distorting local markets.  JS...

Website-friendly impact:

Forest degradation can undermine rural livelihoods by damaging the natural resources and ecosystem services on which communities depend.

10. Loss of ecosystem services

A forest provides much more than timber. These benefits are known as **ecosystem services**.

The Sierra Madre's forests provide services such as:

- **Water regulation**
- **Carbon storage**
- **Soil stabilization**
- **Flood regulation**
- **Wildlife habitat**
- **Climate regulation**
- **Pollination and ecological processes**
- **Forest resources for communities**

When forests are degraded, several of these services can decline simultaneously.

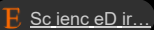
This makes illegal logging an environmental issue as well as a **water, agricultural, economic, and disaster-risk issue**.

11. Forest fragmentation makes recovery more difficult

Deforestation does not always remove an entire forest at once. Logging can create smaller, disconnected patches of remaining forest.

This process is called **forest fragmentation**.

Fragmented forests have more exposed edges and less continuous habitat. This can interfere with wildlife movement and make remaining forest patches more vulnerable to additional human disturbance.

Research examining land-use change in the Northern Sierra Madre Natural Park highlights the importance of protecting remaining forest areas from continuing land-use pressures. 

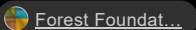
Website-friendly impact:

Fragmentation breaks continuous forests into smaller isolated areas, making ecosystems more vulnerable to further degradation.

12. Illegal logging can reinforce other forms of environmental degradation

Illegal logging can also open previously inaccessible forest areas.

Once roads, trails, and access routes are created, other activities—including agricultural expansion, hunting, charcoal production, mining, and additional timber extraction—can become easier.

Research on the Southern Sierra Madre identified **kaingin, infrastructure expansion, timber poaching, small-scale mining, charcoal making, and natural hazards** among factors associated with forest degradation. 

This means illegal logging can become part of a **larger cycle of forest degradation** rather than being an isolated activity.

The chain reaction

You could present the issue on your website as a simple environmental chain:

Illegal Logging

Loss of Trees and Forest Cover
Habitat Destruction + Soil Exposure
Erosion + Increased Runoff
Landslides + River Sedimentation + Flooding

Watershed Degradation Damage to Agriculture, Infrastructure, Wildlife, and Communities

At the same time:

Forest Loss

Less Carbon Storage
Reduced Climate-Regulation Capacity

And:

Forest Fragmentation

Loss of Wildlife Habitat
Declining Biodiversity
Greater Ecosystem Vulnerability

Key facts you can highlight

Issue	Effect on Sierra Madre
 Forest loss	Reduces forest cover and ecosystem services
 Bio d i v e r s i t y	Destroys and fragments wildlife habitat
 Heavy rainfall	Increases runoff from degraded slopes
 Landslides	Greater vulnerability on exposed, steep slopes
 Flooding	Increased runoff can contribute to downstream flooding
 Watersheds	Increased erosion and sedimentation degrade water syst e m s
 Soil	Loss of vegetation increases soil erosion

Issue	Effect on Sierra Madre
 Typhoons	Degraded ecosystems provide less protection against storm impacts
 Climate	Reduced carbon-storage capacity
 Communities	Threatens natural resources and rural livelihoods
 Other activities	Can facilitate agricultural expansion, mining, hunting, and further forest degradation

A particularly useful statistic

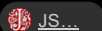
A recent **2026 GMA News report**, citing a Haribon Foundation study, reported an estimated **9,000 hectares of Sierra Madre forest cover lost annually** to a combination of illegal logging/timber poaching, mining, quarrying, kaingin, and development. The same report cited an estimated **130,000 hectares of forest cover lost from 2003–2020**.

Because these figures come from a secondary news report and combine several causes of forest loss, they should be presented with that attribution rather than described as illegal logging alone. 

For a website, I would phrase it as:

According to a 2026 report by GMA News citing the Haribon Foundation, the Sierra Madre has been losing an estimated 9,000 hectares of forest cover annually due to multiple pressures, including illegal logging and timber poaching, mining, quarrying, kaingin, and development.

Recommended references for your website

- **Climate Change Commission — “Sierra Madre: Mountain Range for Resilience” (2024)**: useful for the mountain range's role in climate resilience, biodiversity, carbon storage, and protection from tropical cyclones. 
[Climate Change Commission article](#)
- **Van der Ploeg et al. — “Illegal Logging in the Northern Sierra Madre Natural Park, the Philippines” (Conservation & Society, 2011)**: particularly useful because it directly examines illegal logging in the Northern Sierra Madre and its effects on biodiversity, livelihoods, ecosystems, and sustainable development. 
[JSTOR article](#)
- **Forest Foundation Philippines — Southern Sierra Madre Deforestation and Forest Degradation Analysis**: useful for more recent evidence from General Nakar, Quezon and Rodriguez, Rizal, including timber poaching, kaingin, mining, charcoal

making, and infrastructure expansion.  Forest Foundat...

Forest Foundation Philippines study

- **GMA News — Sierra Madre and nature-based solutions (2026):** useful for the recent forest-loss figures and explanation of the relationship between forests, flooding, and natural hazards.  GMA Netw...

GMA News article

- **iForest — Illegal forest activity in southeastern Sierra Madre (2022):** useful for academic evidence concerning illegal logging, slash-and-burn farming, agricultural expansion, biodiversity, and protected areas.  iforest.sisef....+1

iForest research article

For a school/environmental website, the strongest structure would be: *What is happening → Causes → Effects on biodiversity → Effects on water/flooding → Effects on communities → Climate effects → What can be done → References.* That keeps the page informative without making unsupported claims that every flood or landslide in the region is caused by illegal logging.